Submodule 38.08: Multi-Class Classification ANN: Problem Formulation aur Architecture (Hinglish Guide)

1. Problem Samajhna: Multi-Class Classification

Pichle submodules mein humne **Regression** seekha jahan model continuous numeric value (jaise Power Output MW) predict kar raha tha. Is submodule mein hum **Multi-Class Classification** implement karenge jahan target variable discrete category hota hai:

$$y \in \{0, 1, 2, \ldots, C-1\}$$

Hamara project dataset hai **Date Fruit Dataset** (`DateFruit_Dataset.csv`):

- **Total Samples**: 898 khajur (date fruit specimens)
- **Features**: 34 physical features (Area, Perimeter, Major Axis, Minor Axis, Solidity, RGB Color moments, etc.)
- **Output Classes**: 7 alag-alag khajur ki varieties:
    1. `BERHI`
    2. `DEGLET`
    3. `DOKOL`
    4. `IRAQI`
    5. `ROTANA`
    6. `SAFAVI`
    7. `SOGAY`

---

2. Regression vs Multi-Class Classification ANN Mein Antar

Feature	Regression Model	Classification Model
Input Shape	4 features	34 features
Output Layer Neurons	**1 neuron** (continuous scalar)	**7 neurons** (har class ke liye ek score/logit)

Feature	Regression Model	Classification Model
Output Activation	None (Linear)	Model ke andar None (`CrossEntropyLoss` Softmax khud lagata hai)
Loss Function	`nn.MSELoss()`	`nn.CrossEntropyLoss()`
Target Tensor Type	`torch.float32`	`torch.long` (Integer 0 to 6)
Evaluation Metric	MSE, RMSE, R^2 Score	Accuracy Percentage ($\%$)

---

3. Architecture Design Rules (Neurons aur Layers Chunne Ka Formula)

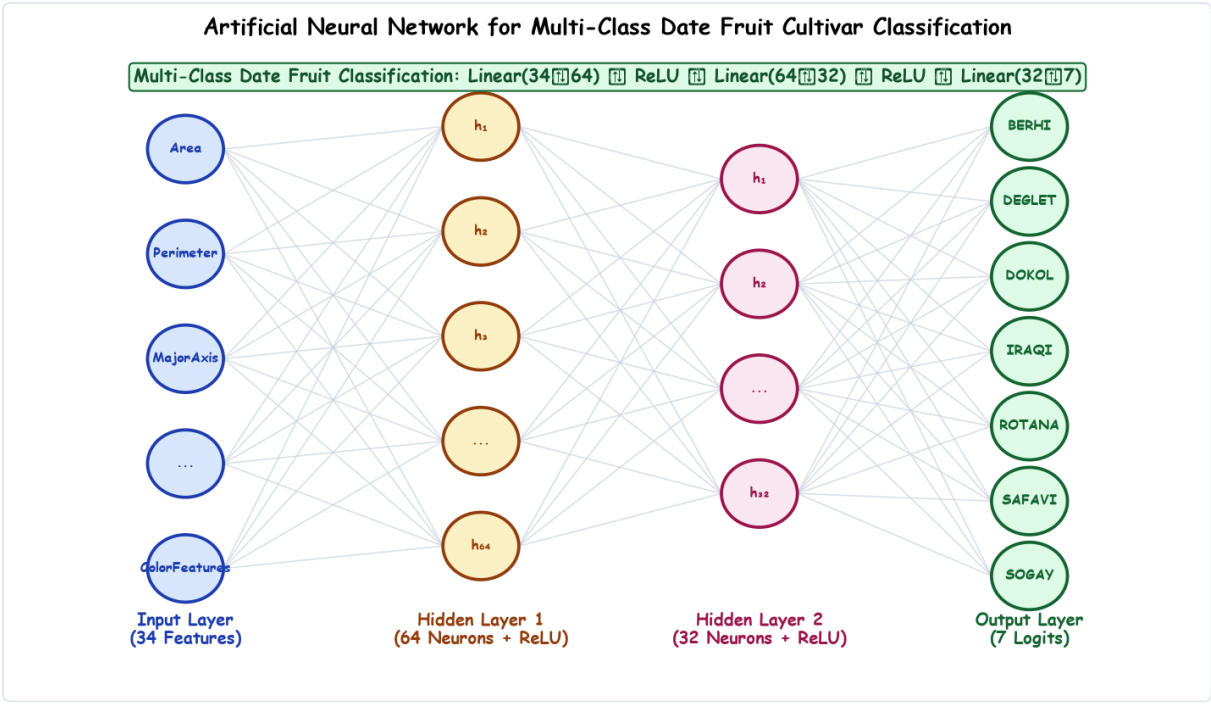
Tabular structured data ke liye Deep Learning best practices:

1. **Hidden Layers Ki Sankhya**: Tabular data ke liye 1 se 2 hidden layers kaafi hoti hain. Jyada deep network banane se overfitting hone lagti hai.
2. **Neurons Ki Sankhya**: General convention ke mutabiq powers of 2 ($16, 32, 64, 128$) use karte hain taaki GPU aur CPU memory optimization best rahe.
3. **Selected Architecture**:
    - Input Layer: 34 input neurons
    - Hidden Layer 1: 64 neurons + ReLU
    - Hidden Layer 2: 64 neurons + ReLU
    - Output Layer: 7 neurons (7 classes ke raw logits)

---

4. Hand-Drawn Educational Diagram: Classification Architecture

Neeche diya gaya clean, student notebook style diagram 34 inputs se lekar 7 class logits tak ke mapping ko visualize karta hai:



---

5. Python Exploration Code

```
import pandas as pd
import numpy as np

# Dataset load karein
df = pd.read_csv("DateFruit_Dataset.csv")
print("Dataframe shape:", df.shape) # 898 rows, 35 columns

# Classes count karein
print(df["Class"].value_counts())

X = df.drop("Class", axis=1)
y = df["Class"]

print("Number of Features:", X.shape[1]) # 34 features
print("Number of Classes :", len(y.unique())) # 7 classes
```